NavTools: Assistive Gaze Tracking & Voice Assistant

Hands-Free Neural-Interface-Like Assistive System

1. What is NavTools?

NavTools is a completely non-invasive, premium assistive control suite that empowers individuals with physical impairments to operate computers hands-free. The system merges webcam-based eye gaze tracking with an offline voice-activated assistant named 'Jim'.

Unlike traditional biosignal-based interfaces (like EEG/EOG) that require extra hardware on the body, NavTools requires ZERO extra hardware. It leverages standard computer webcams and microphones to deliver smooth cursor movement, rich facial-gesture-based clicking, and highly responsive voice command execution.

2. Key Features

- Gaze Control: Uses MediaPipe Face Mesh landmark tracking for accurate cursor positioning.
- Gestures: Blink and Wink gesture recognition for clicks (Left, Right, Double, Click and Drag).
- Voice Command: 80+ pre-configured offline/online speech macros powered by the 'Jim' Assistant.
- Configuration: Unified Tkinter dark-mode settings panel for calibration and adjustments.
- Attention Gating: Thread-safe inter-module synchronization prevents accidental clicks when looking away.
- TTS System: High-fidelity Kokoro ONNX voice generation with local SAPI5 fallback for offline speech.

3. How Does It Work?

The system uses standard input devices (Microphone and Webcam):

1. MediaPipe Landmark Engine:
   - Analyzes video frames to track your eye pupils to move the cursor (X, Y).
   - Detects Eye Aspect Ratio (EAR) to register blinks and winks for mouse clicks.

2. Speech Recognition & Command Engine:
   - Listens for wake words ('Hey Jim', 'Wake up Jim') and parses spoken commands.

3. Thread-Safe State Synchronization:
   - An Attention Gate pauses commands and cursor movement if you are not looking at the screen.

4. Output Interfaces:

- Emulates mouse and keyboard actions via PyAutoGUI.
- Gives verbal feedback using Kokoro ONNX or Windows SAPI5 Speech.

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4. Technologies Used

This project was built entirely using open-source technologies:

Category	Technology / Hardware
Hardware Requirements	Standard Webcam and Microphone (No extra gear)
Programming Language	Python 3.9+
Computer Vision	OpenCV, MediaPipe (Face Mesh)
Voice Assistant	Kokoro-ONNX (TTS), SpeechRecognition, PyAudio, pywin32
User Interface	Tkinter (Python GUI)
System Automation	PyAutoGUI, PyGetWindow

5. Setup & How to Run

Follow these simple steps to start the application:

Step 1: Install Requirements

Ensure you have installed the required dependencies via 'pip install -r requirements.txt'.

Step 2: Launch the Settings Dashboard

Open a terminal in the project folder and type:

```
python -m src.gui_app
```

Step 3: Interact

A gorgeous, dark-themed GUI control panel will appear. You can configure gaze EMA smoothing, choose camera inputs, toggle live previews, and start the Gaze Tracker or Voice Assistant. Alternatively, you can run 'python -m src.multimodal_launcher' to start both modules immediately.